

```
import java.util.*;
import java.io.*;
import java.time.LocalDate;
import java.time.format.DateTimeFormatter;
```

```
class Student {
    private String name;
    private String studentId;

    public Student(String name, String studentId) {
        this.name = name;
        this.studentId = studentId;
    }

    public String getName() { return name; }
    public String getStudentId() { return studentId; }

    public String toString() {
        return name + " (ID: " + studentId + ")";
    }
}
```

```
class Book {
    private int id;
    private String title;
    private String author;
    private boolean isIssued;
    private LocalDate borrowDate;
    private LocalDate dueDate;
    private Student borrower;

    public Book(int id, String title, String author) {
        this.id = id;
        this.title = title;
        this.author = author;
        this.isIssued = false;
        this.borrowDate = null;
        this.dueDate = null;
        this.borrower = null;
    }

    public int getId() { return id; }
    public String getTitle() { return title; }
    public String getAuthor() { return author; }
    public boolean isIssued() { return isIssued; }
    public LocalDate getBorrowDate() { return borrowDate; }
```

```
public LocalDate getDueDate() { return dueDate; }
public Student getBorrower() { return borrower; }
```

```
public void issueBook(Student s) {
    if (!isIssued) {
        isIssued = true;
        borrower = s;
        borrowDate = LocalDate.now();
        dueDate = borrowDate.plusDays(14);
    }
}
```

```
public void returnBook() {
    isIssued = false;
    borrower = null;
    borrowDate = null;
    dueDate = null;
}
```

```
public String toString() {
    DateTimeFormatter fmt = DateTimeFormatter.ofPattern("dd-MM-yyyy");
    String info = "ID: " + id + ", Title: " + title + ", Author: " + author +
        ", Status: " + (isIssued ? "Issued" : "Available");
    if (isIssued) {
        info += ", Borrowed by: " + borrower +
            ", Borrowed: " + borrowDate.format(fmt) +
            ", Due: " + dueDate.format(fmt);
    }
    return info;
}
```

```
public String toCSV() {
    String s = id + "," + title + "," + author + "," + isIssued;
    if (isIssued) {
        s += "," + borrowDate + "," + dueDate + "," + borrower.getName() + "," +
        borrower.getStudentId();
    }
    return s;
}
```

```
public static Book fromCSV(String line) {
    String[] data = line.split(",");
    Book b = new Book(Integer.parseInt(data[0]), data[1], data[2]);
    if (Boolean.parseBoolean(data[3])) {
        b.isIssued = true;
        b.borrowDate = LocalDate.parse(data[4]);
        b.dueDate = LocalDate.parse(data[5]);
        b.borrower = new Student(data[6], data[7]);
    }
}
```

```
    }  
    return b;  
}  
}
```

```
class Library {  
    private ArrayList<Book> books;  
    private static final String FILE_NAME = "library_data.txt";
```

```
    public Library() {  
        books = new ArrayList<>();  
        loadBooksFromFile();  
    }
```

```
    public void addBook(Book book) {  
        books.add(book);  
        System.out.println("Book added successfully!");  
        saveBooksToFile();  
    }
```

```
    public void showAllBooks() {  
        if (books.isEmpty()) {  
            System.out.println("No books in the library.");  
            return;  
        }  
        for (Book b : books) System.out.println(b);  
    }
```

```
    public void searchByTitle(String title) {  
        boolean found = false;  
        for (Book b : books) {  
            if (b.getTitle().equalsIgnoreCase(title)) {  
                System.out.println(b);  
                found = true;  
            }  
        }  
        if (!found)  
            System.out.println("No book found with title: " + title);  
    }
```

```
    public void issueBook(int id, Student s) {  
        for (Book b : books) {  
            if (b.getId() == id) {  
                if (!b.isIssued()) {  
                    b.issueBook(s);  
                    DateTimeFormatter fmt = DateTimeFormatter.ofPattern("dd-MM-yyyy");  
                    System.out.println("Book issued successfully!");  
                }  
            }  
        }  
    }
```

```

        System.out.println("Borrowed by: " + s.getName());
        System.out.println("Borrow Date: " + b.getBorrowDate().format(fmt));
        System.out.println("Due Date: " + b.getDueDate().format(fmt));
        saveBooksToFile();
    } else {
        System.out.println("Book already issued!");
    }
    return;
}
}
System.out.println("Book ID not found!");
}

```

```

public void returnBook(int id) {
    for (Book b : books) {
        if (b.getId() == id) {
            if (b.isIssued()) {
                b.returnBook();
                System.out.println("Book returned successfully!");
                saveBooksToFile();
            } else {
                System.out.println("This book was not issued!");
            }
            return;
        }
    }
    System.out.println("Book ID not found!");
}

```

```

private void saveBooksToFile() {
    try (BufferedWriter bw = new BufferedWriter(new FileWriter(FILE_NAME))) {
        for (Book b : books) {
            bw.write(b.toCSV());
            bw.newLine();
        }
    } catch (IOException e) {
        System.out.println("Error saving books: " + e.getMessage());
    }
}

```

```

private void loadBooksFromFile() {
    try (BufferedReader br = new BufferedReader(new FileReader(FILE_NAME))) {
        String line;
        while ((line = br.readLine()) != null) {
            books.add(Book.fromCSV(line));
        }
    } catch (FileNotFoundException e) {
    } catch (IOException e) {
        System.out.println("Error loading books: " + e.getMessage());
    }
}

```

```
}  
}
```

```
public class LibraryManagementSystem {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        Library library = new Library();  
        int choice;  
  
        do {  
            System.out.println("\n===== Library Management System =====");  
            System.out.println("1. Add Book");  
            System.out.println("2. Show All Books");  
            System.out.println("3. Search Book by Title");  
            System.out.println("4. Issue Book");  
            System.out.println("5. Return Book");  
            System.out.println("6. Exit");  
            System.out.print("Enter your choice: ");  
            choice = sc.nextInt();  
            sc.nextLine();  
  
            switch (choice) {  
                case 1:  
                    System.out.print("Enter Book ID: ");  
                    int id = sc.nextInt();  
                    sc.nextLine();  
                    System.out.print("Enter Title: ");  
                    String title = sc.nextLine();  
                    System.out.print("Enter Author: ");  
                    String author = sc.nextLine();  
                    library.addBook(new Book(id, title, author));  
                    break;  
  
                case 2:  
                    library.showAllBooks();  
                    break;  
  
                case 3:  
                    System.out.print("Enter Title to search: ");  
                    String searchTitle = sc.nextLine();  
                    library.searchByTitle(searchTitle);  
                    break;  
  
                case 4:  
                    System.out.print("Enter Book ID to issue: ");  
                    int issuelid = sc.nextInt();  
                    sc.nextLine();
```

```
System.out.print("Enter Student Name: ");
String stuName = sc.nextLine();
System.out.print("Enter Student ID: ");
String stuld = sc.nextLine();
library.issueBook(issueId, new Student(stuName, stuld));
break;
```

case 5:

```
System.out.print("Enter Book ID to return: ");
int returnId = sc.nextInt();
library.returnBook(returnId);
break;
```

case 6:

```
System.out.println("Exiting... Data saved successfully!");
break;
```

default:

```
System.out.println("Invalid choice! Try again.");
}
```

```
} while (choice != 6);
```

```
sc.close();
```

```
}
}
```